

HMDA 2024

RESPONSIBLE ML CAPSTONE

XGBoost Mortgage Approval Model

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MAIN REPORT QUESTION

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x x x
x x x
x x x
x x x



What is the **optimization objective**?



What are the known **failure modes**?



How are **subgroup errors** measured?



What **risks** remain after mitigation?



Is **deployment defensible**?



What is the optimization objective?

Binary classification task: Approve or Deny a mortgage application

Filtered data
4,354,586
rows

Component	Our Choice	Rationale
Main optimization metric	AUC	The dataset: Approvals > Denials. AUC better captures ranking quality.
Training objective	Log-loss	The quality of predicted approval probabilities.
Fairness threshold	AIR < 0.80, FPR, FNR	Flags potential disparate impact.

Stakeholders

Applicants	Lenders / Loan Review Teams	Protected / Minority Groups
Avoid false denial and unfair credit loss	Reliable approval probability ranking	Detect lower selection rates or higher error rates

Features

- **Numeric_features**
property_value
income
tract_population
ffiec_msa_md_median_family_income
tract_to_msa_income_percentage
- **Categorical_features**
state_code
derived_loan_product_typed
erived_dwelling_category
loan_purpose
lien_status
occupancy_type





What is the optimization objective?

- XGBoost achieved the strongest overall ranking and probability performance.
- Random Forest had lower FNR, but weaker AUC/log-loss and higher FPR.
- XGBoost serves as the main model in our audit.

Model	Accuracy	AUC	Log Loss	Precision	Recall	FPR	FNR
Logistic Regression	0.74	0.74	0.51	0.76	0.93	0.77	0.05
XGBoost	0.78	0.80	0.46	0.80	0.93	0.60	0.06
Random Forest	0.76	0.78	0.49	0.77	0.96	0.76	0.03

XGBoost Confusion Matrix		
Actual Label	Denied (0)	Approved (1)
	Denied (0)	Approved (1)
Denied (0)	TN 103,647	FP 165,839
Approved (1)	FN 47,345	TP 660,470





What are the known failure modes?

Failure Type	Trigger	Affected Group	Severity
Proxy Discrimination	Tract income features in SHAP top 10 (flagged as proxies)	Minority applicants	High
Adversarial Gaming	Loan purpose and income are top SHAP drivers; both are known in HMDA applications	All applicants	High
Feature Drift	AUC drops from 0.68 to 0.56 across prediction buckets	Borderline applicants	High
Fairness Disparity	Black AIR = 0.93; higher FNR vs White Applicants	Black Applicants	High



How do the failure modes appear in practice?

Manipulation Risk

- Income and loan purpose drive predictions
- Susceptible to probing and strategic input changes

Fairness Risk

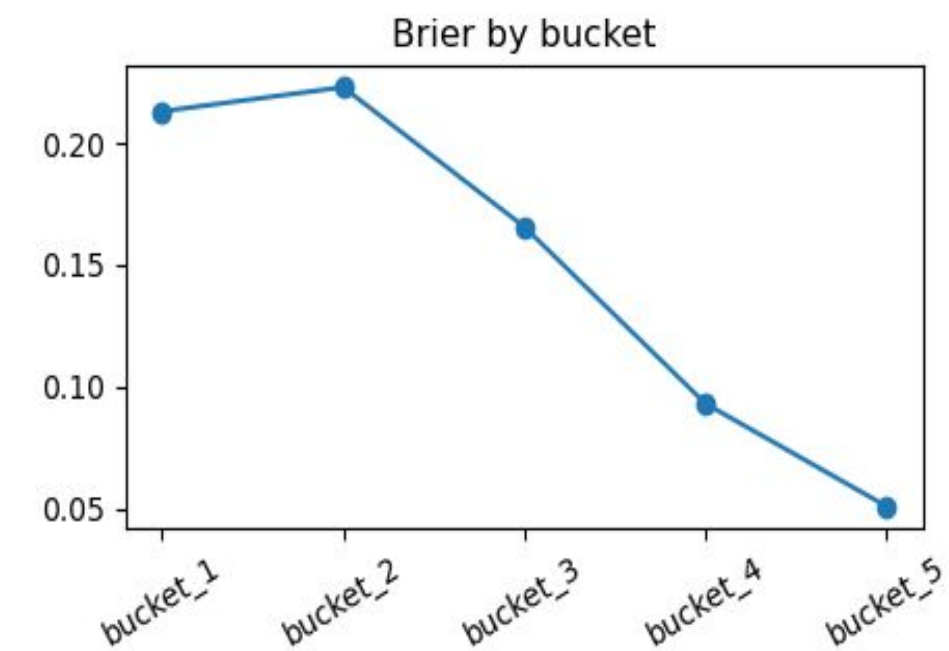
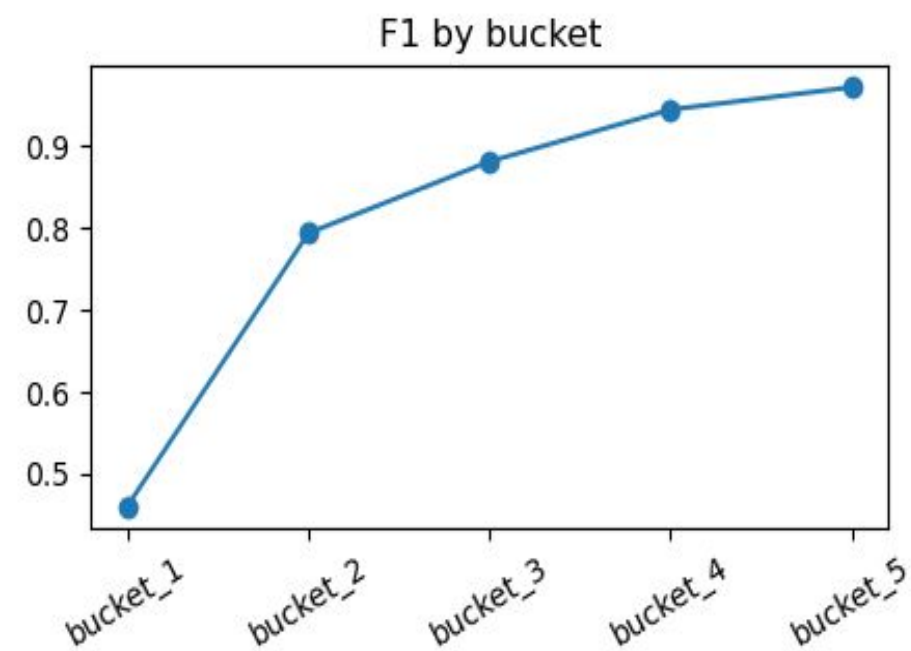
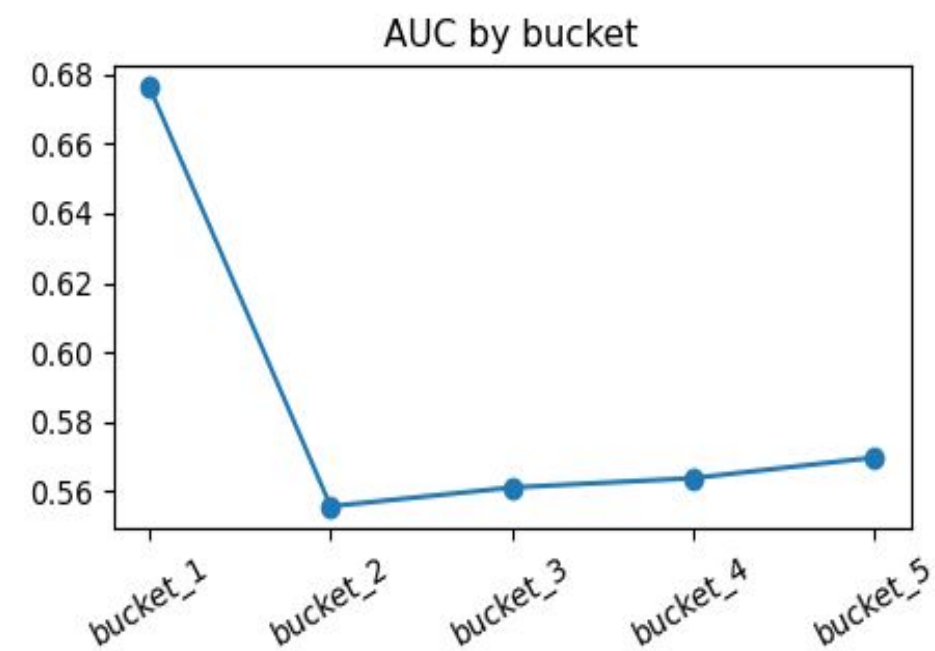
- Tract-level features act as demographic proxies
- Creates disparity without using race directly

Boundary Weakness

- Model performs worst near decision threshold
- Errors concentrated on borderline applicants

Limited Actionability

- Counterfactuals require unrealistic changes
- Denied applicants cannot act on model feedback





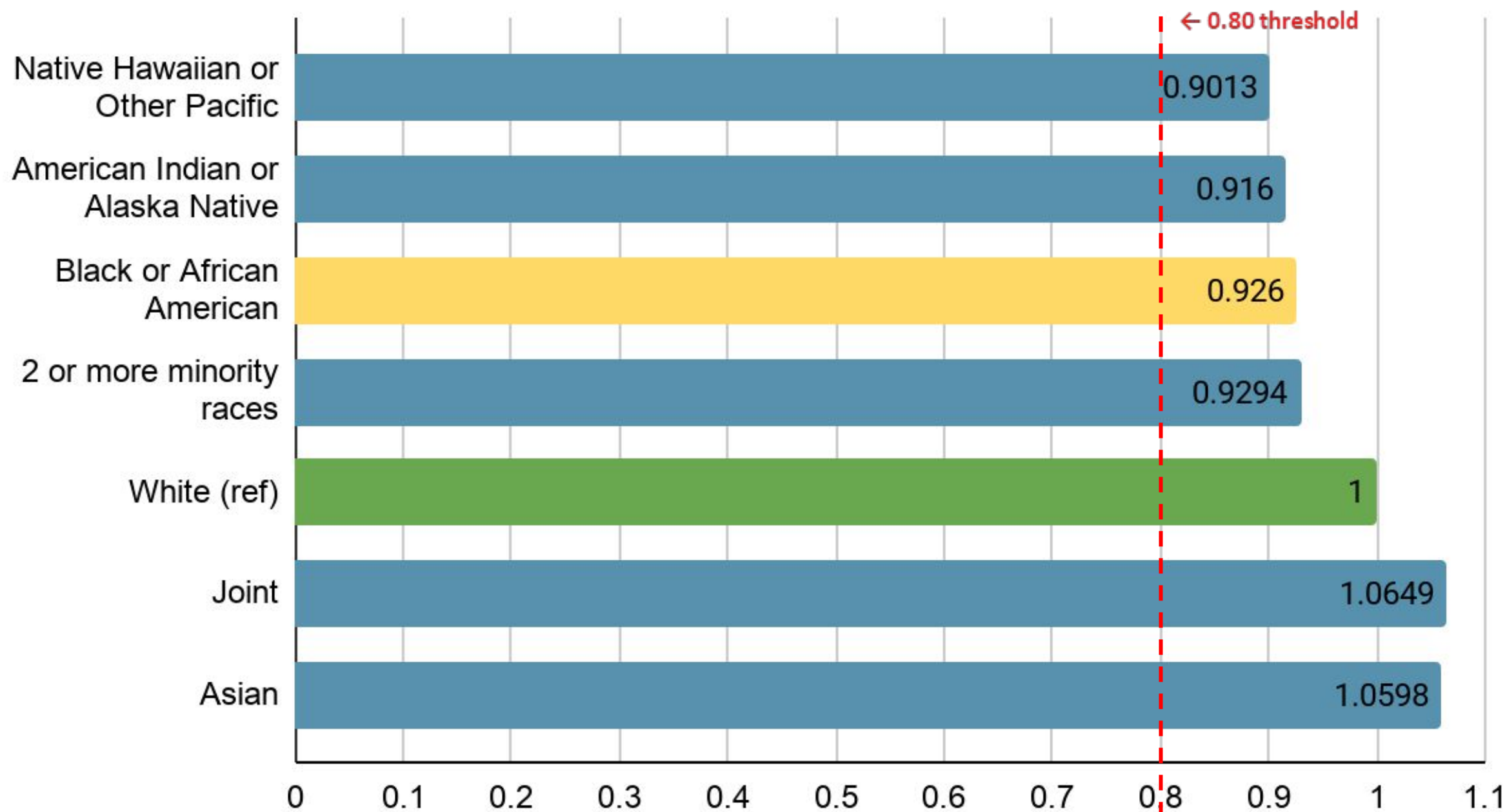
How are subgroup errors measured?

AIR Adverse Impact Ratio	ME Marginal Effect	SMD Standardized Mean Difference
Approval rate of a protected group ÷ Approval rate of the reference group	Absolute difference in selection rates between a group and the reference group	Gap in predicted probability scores between groups, measured in standard deviations .
Reference groups: White, Male, Not Hispanic or Latino	Captures absolute gap in percentage points	Gap in predicted probability scores, measured in standard deviations
Four-Fifths Rule: $AIR < 0.80 \Rightarrow$ potential fair-lending violation	Negative = disadvantaged group	Cohen's d: small <0.2 , medium <0.5 , large <0.8



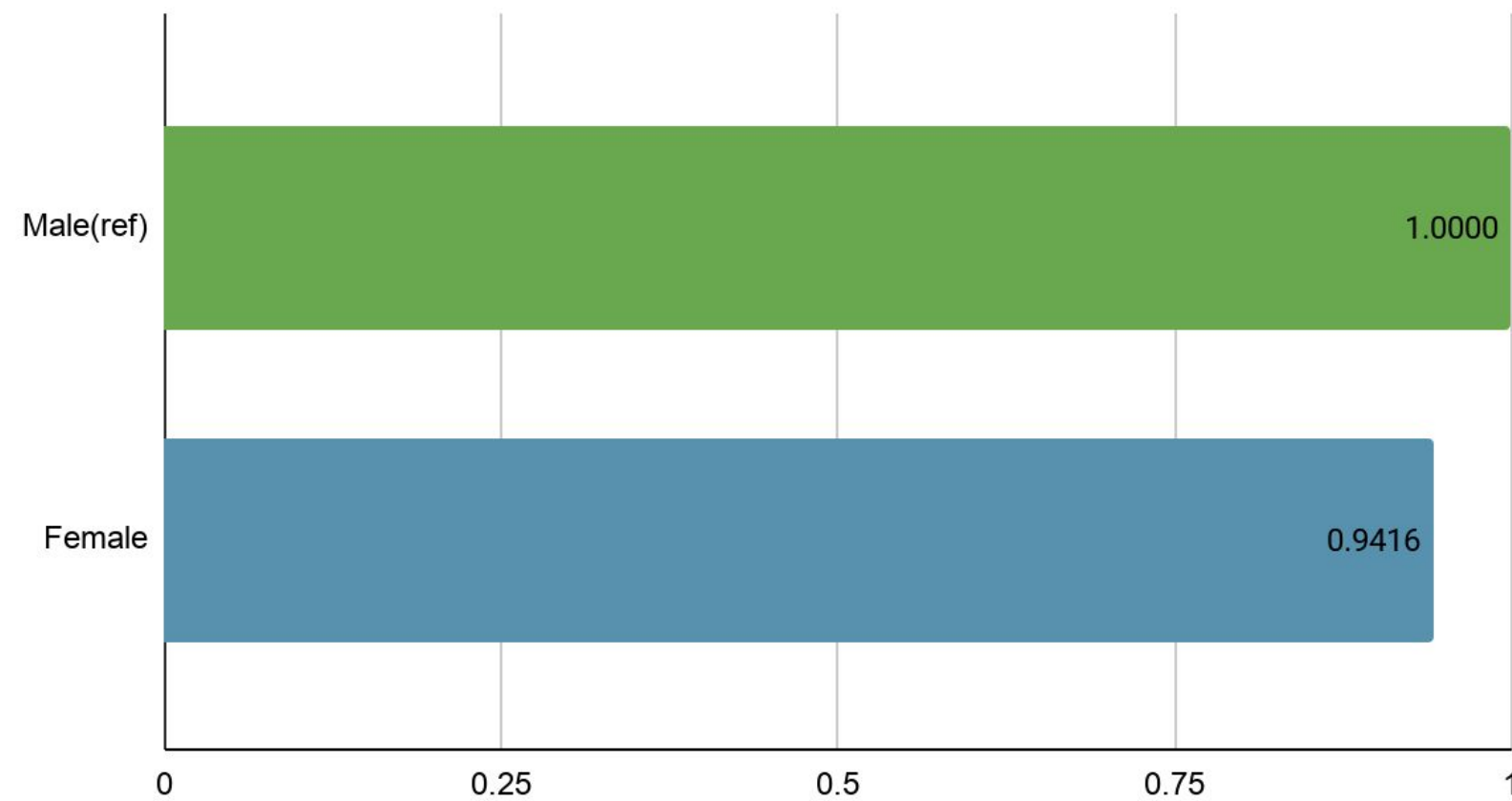
How are subgroup errors measured?

AIR by Race (reference group: White)

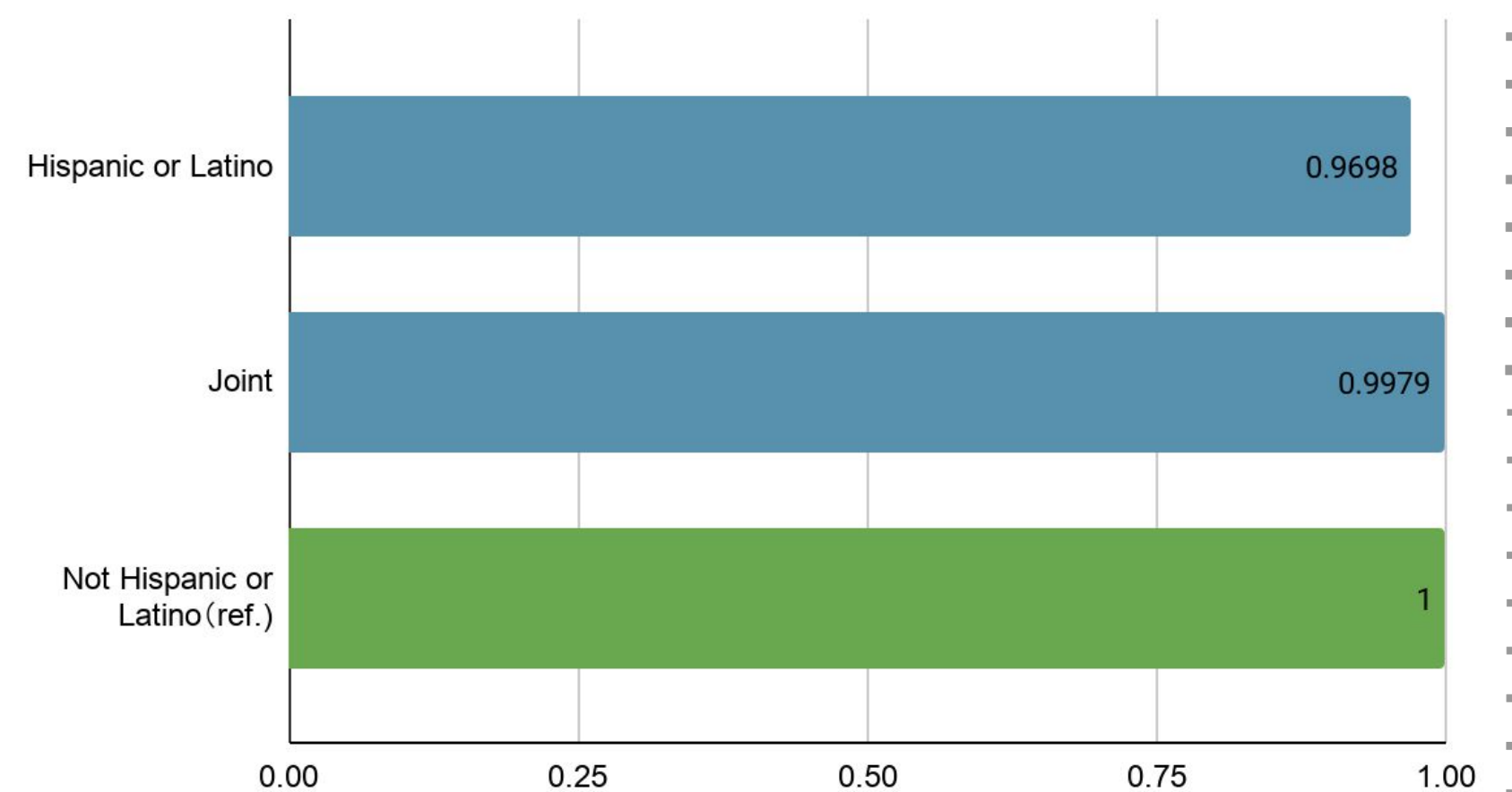


How are subgroup errors measured?

AIR by Sex (reference group: Male)



AIR by Ethnicity (reference group: Not Hispanic or Latino)

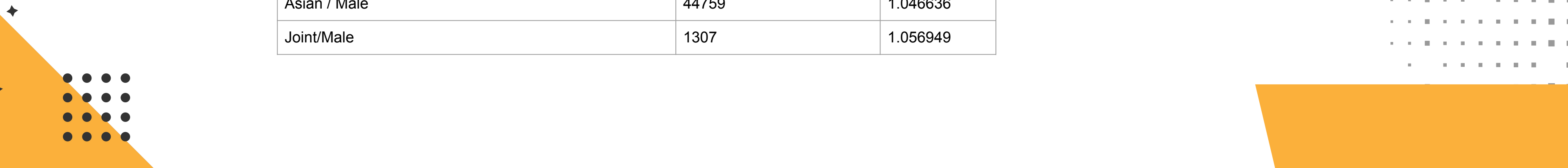




How are subgroup errors measured?

AIR by Race × Sex Subgroup (reference group: White/Male)

subgroup	n	AIR
American Indian or Alaska Native / Female	3870	0.846754
Native Hawaiian or Other Pacific Islander / Female	1126	0.838682
2 or more minority races /Female	1444	0.884528
Black or African American / Female	60695	0.878563
Native Hawaiian or Other Pacific Islander / Male	1963	0.907933
American Indian or Alaska Native / Male	6016	0.929422
2 or more minority races / Male	1762	0.931177
Black or African American / Male	54397	0.938463
White / Female	248414	0.946693
White/Male	391286	1.000000
Joint /Female	1084	1.025774
Asian / Female	23043	1.020922
Asian / Male	44759	1.046636
Joint/Male	1307	1.056949





The remaining risks

Intersectional Fairness Sensitivity	Performance Drift Across Score Segments	Uneven Robustness for Vulnerable Groups
- Intersectional groups (e.g., combinations of race and sex) show fairness sensitivity that isn't visible when looking at race-only or sex-only metrics. - Indicates fairness outcomes can shift quickly for certain subpopulations, even when overall metrics appear stable. - Mitigation: Build intersectional fairness checks into routine monitoring and apply targeted calibration where disparities persist.	- The model behaves differently across parts of the score range, making outcomes less consistent for applicants near key decision thresholds. - This drift can amplify fairness sensitivity, since some groups cluster in the more unstable score regions. - Mitigation: Monitor score-bucket performance and apply threshold-level calibration to stabilize behavior.	- Some applicant segments show reduced stability when inputs shift, meaning small changes in their profiles can lead to noticeably different outcomes. - This creates higher volatility for borderline applicants in these groups, even when overall model performance looks steady. - Mitigation: Strengthen slice-level monitoring and apply targeted calibration or guardrails to ensure more consistent treatment.

Is deployment defensible?

Deployment Decision CONDITIONAL DEPLOY	Monitoring Plan	Risk Acceptance
Why deploy: XGBoost is our strongest model – AUC 0.7977, accuracy 0.7819. The gap between training and test performance is just -0.0016, which tells us the model has learned real patterns, not noise. It held up under three economic stress scenarios and shows no signs of data drift. The top decision drivers – loan purpose, income, and property value – are financially sound and explainable to regulators. Conditions: This model must operate as a decision-support tool only. A human must review every final decision. Security controls must be in place before go-live.	Performance (monthly): Track AUC, FPR, and FNR. If AUC drops below 0.75 or FPR rises above 0.70, the model is pulled for retraining. Fairness (quarterly): Monitor approval rates across race, sex, and ethnicity against our 0.80 policy threshold. No group currently falls below it, but two intersectional groups are close and will be watched – American Indian or Alaska Native females (AIR 0.847) and Native Hawaiian or Pacific Islander females (AIR 0.839). Drift (monthly): Track whether incoming data still matches what the model was trained on using PSI. If any feature exceeds 0.25, retrain immediately. All features are currently stable. Security (weekly) : Review API logs for unusual query patterns that may signal gaming or model extraction attempts.	Accepted: Generalization gap of -0.0016, stable PSI across all features, and graceful degradation under stress testing support conditional deployment. Accepted with Conditions: No group currently breaches the 0.80 AIR threshold. Two intersectional subgroups are close to the boundary and are accepted conditionally, subject to quarterly review and regulatory disclosure. Geographic income proxy variables are flagged and monitored. Not Accepted: Sole automated decision-making without human review. Rate limiting, query logging, input validation, and pipeline integrity checks are non-negotiable before go-live.

Audit Record

Every finding, threshold, and decision in this presentation is documented in our GitHub repository – the formal audit record for this model.



THANK YOU

